

Maxwell Lajoie

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EDUCATION

University of Notre Dame

Bachelor of Science

Major: **Computer Science, Statistics** | Minor: Theology

Notre Dame, IN

May 2027

PROJECTS

AI TENNIS MATCH PREDICTOR

- Developed a tennis match outcome predictor using **Python**, **Pandas**, and **Scikit-learn**, achieving an **83%** accuracy rate with a decision tree model.
- Engineered custom player statistics, including Elo ratings, surface-specific performance, head-to-head comparisons, and win rates over varying game spans, to enhance predictive accuracy.
- Designed and implemented a data preprocessing pipeline to clean and structure CSV datasets, ensuring usefulness of data
- Built and trained a decision tree classifier using Scikit-learn, optimizing feature selection and model parameters for improved performance.

MONTE CARLO PRICING MODEL

- Developed program in **Python** that compares the result of a Monte Carlo pricing model to a Black-Scholes pricing model.
- Simulated stock price movement using the Geometric Brownian Motion model utilizing **matplotlib.pyplot**.
- Utilized **NumPy** and **scipy.stats** to calculate the call of a stock given specified parameters through implementation of the Black-Scholes formula.
- Estimated the call option price using Monte Carlo Simulation and compares with the Black-Scholes price.

AI DISTRACTED DRIVER DETECTION

- Developed a computer vision system using **Python**, **OpenCV**, and **TensorFlow** to detect distracted drivers with **87%** accuracy.
- Collaborated on data preprocessing, **CNN** model design, and hyperparameter tuning to optimize detection performance.
- Used a convolutional neural network (**CNN**) trained on driver behavior datasets, incorporating data augmentation to improve accuracy.
- Achieved model validation through confusion matrix, precision, recall, and F1-score metrics.

PERSONAL SHELL

- Developed a custom command-line shell with **Python** with built-in support for `cd`, `pwd`, `echo`, `type`, and execution of external commands.
- Designed a prompt system that dynamically updates based on the current working directory and user environment.
- Utilized **os**, **shutil**, and **sys** modules to manage system operations and execute commands efficiently.

ACTIVITIES / WORK

HUNGRY ATHLETES

Co-Founder

Garland, TX

Nov 2022 – Present

- Co-founded a nonprofit organization with my brother, partnering with a local food pantry to deliver essential groceries to **23** people in need.
- Coordinated regular food deliveries, ensuring consistent support for underserved populations.
- Created a program to maximize efficiency of delivery routes in **Python** based on the traveling salesman problem

DREAMCATCHERS

Chapter President

Garland, TX

Dec 2020 – July 2023

- Established and led a nonprofit chapter dedicated to fulfilling end-of-life dreams for hospice patients.
- Built partnerships with local hospices, securing participation and expanding the network of care providers.
- Spearheaded fundraising initiatives, successfully raising funds to support dream fulfillment activities.
- Managed the chapter's budget, ensuring efficient allocation of resources to maximize impact.

ADDITIONAL

Technical Skills: Python, SQL, JavaScript, Java, C, C++, MATLAB, R, HTML/CSS, Flutter, Excel

Relevant Coursework: Discrete Math, Linear Algebra & Diff. Equations, Data Structures, Logic Design, Systems Programming

Relevant Libraries: Pytorch, scikit-learn, Pandas, TensorFlow, Matplotlib, os, sys, NumPy, scipy.stats